

Shashank Yelagandula

Data Analyst

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SUMMARY

- Accumulated as a Data Analyst with around 3 years of experience in data mining, cleaning, and visualization using Python (Pandas, NumPy), SQL, Tableau, and Power BI delivering insights from datasets exceeding 30,000 records.
- Knowledgeable in ETL pipelines, big data processing (Azure Synapse, Databricks, Apache Spark, Kafka), and workflow automation to optimize efficiency and data accuracy.
- Proficient in data governance, compliance (GDPR), predictive modeling, A/B testing, and hypothesis testing, ensuring high-quality data for strategic initiatives.
- Knowledge in managing databases (MySQL, PostgreSQL, Oracle), and leverage Jira, Confluence, and Git for seamless project execution.

EDUCATION

GEORGE MASON UNIVERSITY

Master's in Data Analytics Engineering

Fairfax, Virginia
Jan 2023 – Dec 2024

INSTITUTE OF AERONAUTICAL ENGINEERING

Bachelor's in Electronics and Communication Engineering

Hyderabad, India
Sep 2017 – May 2021

PROFESSIONAL EXPERIENCE

MERCK

VA, USA

Data Analyst

Mar 2025 – Current

- Analyzed large-scale healthcare datasets including electronic health records (EHRs), claims, and clinical trial data to identify patient care trends, treatment efficacy, and cost drivers, directly supporting strategic decision-making in pharmaceutical R&D and healthcare operations.
- Developed and optimized SQL queries to extract, aggregate, and join complex data across multiple relational databases, accelerating clinical and operational reporting processes by 35%.
- Built and maintained interactive dashboards and reports using Power BI and Excel (pivot tables, slicers, macros) to visualize patient outcomes, medication adherence, and healthcare resource utilization.
- Performed data cleansing, normalization, and transformation using Python (Pandas, NumPy) to ensure accuracy and consistency across analytical workflows, supporting high-integrity outcomes in regulatory and compliance reporting.
- Conducted root cause analysis and exploratory data analysis (EDA) on process inefficiencies, identifying opportunities to optimize trial workflows and reduce delays in drug development timelines.
- Participated in cross-functional sprint cycles using Agile methodology, contributing to requirement gathering, sprint planning, and delivery of analytics features for real-time clinical monitoring platforms.

ONE COMMUNITY INC

San Gabriel, CA

Data Analyst

Feb 2025 – Feb 2025

- Collected, cleaned, and analyzed survey data to identify key trends and insights for community development initiatives, ensuring accurate and meaningful reporting.
- Created data visualizations and reports using Excel and Tableau, helping stakeholders understand the impact of programs and make data-driven decisions.

TATA CONSULTANCY SERVICES

Bangalore, India

System Engineer/ Data Analyst

Jun 2021 – Jan 2023

- Developed 15+ interactive Tableau dashboards, improving data accessibility for over 200 stakeholders and decreasing report time by 8–10 hours per week through improved UI/UX, interactive filters, and calculated fields.
- Managed 5+ TB of SQL databases, refining queries and indexing to cut data retrieval time from 20 seconds to under 12 seconds for stored procedures to minimize execution time from 15 seconds to 10 seconds.
- Designed & automated 10+ ETL pipelines in Azure Data Factory, integrating structured & unstructured data sources (Azure Data Lake, SQL Server, REST APIs), reducing manual workload by 30+ hours per month.
- Built 10+ ETL pipelines using Azure Data Factory, integrating structured and unstructured data from Azure Data Lake, SQL Server, and REST APIs, saving 30+ hours of manual work monthly.
- Automated file transfers and batch processes using Stonebranch, GoAnywhere, and Connect Direct, ensuring secure data exchange across banking and financial platforms for PSCU BFSI while maintaining compliance.
- Automated Excel-based reports and templates using Power Query, VBA, and Python (openpyxl, xlrd, Pandas), saving over 25 hours per month in manual reporting tasks.

- Streamlined Python-based automation scripts for data cleansing, transformation (reshaping datasets up to 10 million records), and validation (schema enforcement, business rule checks), improving data reliability.
- Applied NumPy for advanced numerical operations, time-series forecasting, and matrix transformations, accelerating data processing by 5–7 minutes per large dataset.
- Formulated forecasting models using Excel (Regression, Time Series), Pandas, and NumPy, improving prediction accuracy for customer behaviour and financial performance, aiding in quarterly budget planning.
- Harnessed Tableau dashboards by refining UI/UX, interactive filters, calculated fields, and leveraging extracts & indexing, reducing dashboard load times from 15 seconds to under 8 seconds.
- Designed & maintained high-performance SQL stored procedures, supporting real-time business intelligence and reducing query execution time from 12 seconds to 6 seconds.
- Presented insights using Excel (Pivot Tables, Advanced Formulas), Pandas (groupby, pivot), Matplotlib, Seaborn, and PowerPoint, contributing to 15+ executive strategy meetings.

TRIGENT SOFTWARE

India

Data Analyst Intern

Aug 2020 -May 2021

- Refined SQL queries for data filtering, and aggregation, ensuring faster query execution and diminishing time.
- Pruned complex SQL queries to retrieve and manipulate data from relational databases, enhancing the speed and accuracy of report generation for timely business insights.
- Systematized data analysis and reporting tasks with Excel models, incorporating advanced functions like Pivot Tables, VLOOKUP, and Macros, automating repetitive tasks and improving weekly workflows.
- Monitored Oracle databases to ensure high availability and reliability, implementing regular backups and disaster recovery procedures to minimize downtime and data loss.
- Secured sensitive data within PostgreSQL databases by applying access controls, encryption, and validation measures, maintaining data integrity and protecting over 1,000 GB of data from unauthorized access.
- Streamlined Pandas for data aggregation and filtering, manipulating large datasets to uncover actionable insights and support business functions.
- Conducted A/B testing to assess the impact of changes on key metrics, providing data-backed recommendations that upgraded conversion rates and user engagement.

PROJECTS

LARGE LANGUAGE MODEL FOR KNOWLEDGE GRAPH EXTRACTION & REASONING BASED ON COMPLEX DATA

- Implemented machine learning pipelines using Amazon SageMaker to transform unstructured data into knowledge graphs with 75% extraction accuracy.
- Deployed the output in a Neo4j graph database within a Docker container on EC2, ensuring efficient knowledge representation & Apache Airflow for automated scheduling real-time updates, boosting processing efficiency by 40%.
- Developed workflows to extract and preprocess 50,000+ PubMed abstracts, reducing manual processing time by 30%. Stored structured data in Amazon S3 for scalability. Built an interactive UI using Streamlit, LangChain, and LLMs.

ZERO-SHOT QUESTION ANSWERING USING PRE-TRAINED MODELS

- Formulated a zero-shot QA system using mBERT and XLM-RoBERTa, reducing retraining time and provided industry-agnostic solutions for faster decision-making across diverse use cases.
- Evaluated performance using the Natural Questions dataset for achieving context-aware answers.

CARDIOVASCULAR HEALTH PATTERNS USING STATISTICAL AND PREDICTIVE ANALYTICS

- Facilitated predictive models using gradient boosting algorithms to achieve 74% accuracy and 81% AUC, uncovering critical health indicators for cardiovascular disease prevention and enhancing healthcare diagnostics.
- Engaged epidemiological methodologies to uncover key predictors of CVD, contributing to healthcare diagnostics and preventive health measures.

TECHNICAL EXPERIENCE

- **Programming:** Python (Pandas, NumPy, Scikit-learn), SQL
- **Project Management:** Jira, Confluence, Git, Agile
- **Data Visualization:** Tableau, Power BI, Matplotlib, Seaborn
- **Databases:** MySQL, PostgreSQL, Oracle SQL, AWS S3, Azure Blob Storage
- **ETL & Data Engineering:** Apache Airflow, Azure Functions, AWS Lambda, GitHub Workflows
- **Automation & Optimization:** Workflow Automation, Process Optimization, Performance Tuning
- **Cloud & Big Data:** Azure Data Factory, Azure Synapse, Databricks, AWS Glue, Apache Spark, Kafka
- **Data Governance & Compliance:** Data Quality, GDPR, Data Governance, Compliance Management
- **Data Analytics & BI:** Data Mining, Data Cleaning, Data Modeling (ERwin), Predictive Modeling, A/B Testing, Hypothesis Testing, Statistical Analysis (SPSS)